

Shubhankar Gahlot

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EDUCATION

Data Science, M.S.

Illinois Institute of Technology Chicago

Product Management for AI & Data Science

365datascience

Design (HCI), B.Des.

Indian Institute of Technology Guwahati

Aug 2016 - May 2018

Dec 2022

Jul 2008 - May 2012

EXPERIENCE

Lead ML & MLOps Architect, Symbiose Management

2025 - Present

- Build scalable ML workflows & MLOps pipeline for various clients in insurance, finance and forest management.

- Work with cross-functional teams to design, develop and deploy ML solutions that solve real-world problems.

Tech Stack: Python, Dask, Ray, Azure Cloud, Terraform, Kubernetes, PostgreSQL, DuckDB, Docker, MLOps pipelines

Co-founder & CTO, SoranoAI

2025

- **Technical Leadership:** Led a 5-person team building scalable ML and MLOps systems for global climate and solar forecasting.

- **Scalable Machine Learning & MLOps:** Architected a distributed climate data processing pipeline using Dask, Ray, and GCP, enabling the processing of 40 TB of ECMWF global climate data in days instead of months. Expanded Kumo, an AI meteorologist, integrating LLMs, MCP tools and Claude agents for automated forecasting.

- **Model Accuracy & Performance:** Achieved 10-15% RMSE improvement in temperature and 5% RMSE improvement in solar power forecasts via large-scale bias correction and ensemble modeling techniques.

- **Product & Business Impact:** Deployed forecasting solutions for enterprise clients; secured Techstars 2025 (\$120k), Stanford StartX, Stanford TomKat Sustainability Sponsorship, and angel funding to scale R&D and climate innovation.

Tech Stack: Python, Dask, Ray, GCP, MCP Tools, Claude LLM, MLOps pipelines

Research Scientist, NASA IMPACT, University of Alabama Huntsville

2020 - 2022

- **Machine Learning Research:** Developed and deployed deep learning technologies for NASA's IMPACT project, focusing on Earth Data accessibility via computer vision and NLP.

- **MLOps, Data Engineering & Software Development:** Built tools for seamless deployment and continuous integration of ML projects.

- **Project Management:** Led an 8+ member team to execute ML and data engineering initiatives, ensuring alignment with NASA objectives. Managed timelines, tasks, and milestones, achieving on-time delivery. Improved transparency through stakeholder communication and optimized budgets by 10%. Applied Agile methodologies to enhance team collaboration and output quality.

Scalable Data Science & Deep Learning Deployment Research

2018 - 2020

Associate, Oak Ridge National Lab, Oak Ridge, TN

- **Data Analysis & Benchmarking:** Conducted analysis to evaluate neural network operations, achieving 87% scaling efficiency for ResNet50 on **6,144 GPUs**; findings published in SC'19.

- **ML Modeling:** Designed a ResNet-based classifier for crystallography space groups (70% accuracy), earning runner-up in the SMC Data Challenge 2019.

- **Software Development:** Built tools to streamline ML framework deployment (TensorFlow, PyTorch), reducing deployment time by 50%. ([gitlab](#))

- **Outreach & Training:** Presented a webinar on building DL libraries for HPC systems, attended by 100+ researchers.

- **Project Management:**

- Led a proof-of-concept dashboard for real-time workload tracking on Summit supercomputer.

- Coordinated efforts for scaling ML models on Summit, overseeing SC'19 presentation and publication.

Sponsored Data Science Practicum (Data Analyst), Prospect

2017

Resources Inc. Chicago, IL

- **Statistical Analysis:** Implemented and tested a buying strategy based on **Moving Average (MACD)** technique against existing regression-based buying strategy which helped in improving the hedging predictions in the electrical and energy sector. Results compiled in a technical report. ([report](#))

• **Software Design:** 3+ years of experience in designing software for multiple clients in India such as WIPRO Technologies, Asian Paints, etc.

• **Product Design & Management:** 2+ years of experience in conceptualizing, designing & delivering software products and handling teams of multiple sizes for various products. One of the products conceptualized and managed *Rutogo* was **acquired** by ixigo.com. ([article](#))

PROJECTS

Eco-Drive Revolution: Reinforcement Learning-Enhanced Cruise Control for Fuel Efficiency and Climate Impact (NeurIPS 2024)

Text to image diffusion model: Trained a U-Net based diffusion autoencoder and CLIP for text input using Fashion MNIST dataset. ([github](#))

Flood extent detection: Designed and trained a multi-modal vanilla U-Net and Feature Pyramid Network models for flood extent segmentation achieving an IOU score of 0.77 ([presentation](#))

Deep learning deploying and scaling strategies: Conducted analytical tests and developed strategies to deploy and scale Deep learning on Summit supercomputer, achieving 87% scaling efficiency on 6,144 GPUs ([IEEE/ACM Workshop](#))

Data optimization for large batch distributed training: Developed a novel data optimization technique for distributed training of neural nets of up to 20k batch size ([DOI](#))

Verb sense disambiguation for densifying knowledge graphs in Earth science ([NASA](#))

Large Language Model BERT-e: Fine-tuned and benchmarked in-house LLM (bert-e) against scibert and bert for extracting Earth Science terms from their definitions. ([huggingface](#))

Towards a Universal Classifier for Crystallographic Space Groups: Developed a computer vision model to classify crystallography space groups that achieved an **accuracy of 70%**. *SMC Data Challenge 2019* runner up ([presentation](#))

Impact of Urban Weather on Energy Use: Conducted statistical analysis on the impact of urban weather on energy use. *SMC Data Challenge 2018* runner up ([presentation](#))

Back-testing hedging strategies: Implemented and back-tested MACD and LSTM based hedging strategies against existing regression-based strategy which improved prediction by 6% ([report](#))

PEER-REVIEWED CONFERENCE & JOURNAL PUBLICATIONS

Gahlot, S. (2024). Eco-Drive Revolution: Reinforcement Learning-Enhanced Cruise Control for Fuel Efficiency and Climate Impact. *Neural Information Processing Systems (NeurIPS), Ver.1*. <https://tinyurl.com/4js2s8b4>

Gahlot, S., Gurung, I., Maskey, M., & Molthan, A. (2022). Flood extent data for machine learning. *NASA-IMPACT, Ver.1*. <https://doi.org/10.24432/C50P62>

Gahlot, S., Kaulfus, A., Priftis, G., & Ramasubramanian, M., et.al. (2022). Time Series Machine Learning Methods for Surface PM2.5 Estimations Using Geostationary Satellites and Numerical Weather Models. *American Meteorological Society Annual Meeting*. <https://ntrs.nasa.gov/citations/20220000473>

Gahlot, S., Gurung, I., Maskey, M., & Ramasubramanian, M., et.al. (2021). Leveraging citizen science and artificial intelligence for monitoring and estimating hazardous events. *American Geophysical Union*. <https://ntrs.nasa.gov/citations/20210025322>

Gahlot, S., Gurung, I., Khatri, M., Maskey, M., & Ramasubramanian, M., et.al. (2021). Application of Artificial Intelligence for Surface PM2.5 Estimations from Geostationary Satellite and Atmospheric Numerical Model Data. *American Meteorological Society*. <https://ntrs.nasa.gov/citations/20205011654>

Gahlot, S., Kaulfus, A., Priftis, G., & Ramasubramanian, M., et.al. (2021). A novel machine learning method for surface PM2.5 estimations from geostationary satellites. *American Geophysical Union*. <https://ntrs.nasa.gov/citations/20210024721>

Gahlot, S., Ramasubramanian, M., Gurung, I., Hänsch, R., Molthan, A., & Maskey, M. (2022). Curating flood extent data and leveraging citizen science for benchmarking machine learning solutions. <https://doi.org/10.1002/essoar.1051103.1>

Bollinger, A., Gahlot, S., Gurung, I., Maskey, M., Ramachandran, R., & Ramasubramanian, M. (2021). Machine learning pipeline for Earth Science using Sagemaker. *American Geophysical Union*. <https://ntrs.nasa.gov/citations/20210024815>

Acharya, A., Davis, C., Gahlot, S., Koehl, D., & Ramasubramanian, M., et.al. (2021). Verb sense disambiguation for densifying knowledge graphs in Earth science. *American Geophysical Union*. <https://ntrs.nasa.gov/citations/20210025330>

Gahlot, S., Shankar, A., & Yin, J. (2020). Data optimization for large batch distributed training of deep neural networks. *Computational Science & Computational Intelligence*. <https://doi.org/10.48550/arXiv.2012.09272>

Dash, S., Gahlot, S., Laanait, N., Maheshwari, K., Morrison, J., Shankar, M., & Yin, J. (2019). Strategies to deploy and scale deep learning on the Summit supercomputer. *Supercomputing*. <https://doi.org/10.1109/DLS49591.2019.00016>

Dash, S., Gahlot, S., Maheshwari, K., Morrison, J., Shankar, A., & Yin, J. (2019). Performance evaluation and best practice recommendations for extreme scale machine learning and deep learning on Summit supercomputer. *AI Expo, Oak Ridge National Lab Postdoctoral Association Research Symposium*.

Gahlot, A., & Gahlot, S. (2019). Changing the state of literacy in the digital age in India. *LINC 2019 Conference, MIT*. <https://doi.org/10.29007/qbpr>

GRANTS & SPONSORSHIP

• Recipient of New Zealand Ministry of Business Innovation and Employment Research funding *University of Canterbury* (35k USD)

WORKSHOP ORGANIZATION	· <u>Enabling Analysis in the Cloud Using NASA Earth Science Data</u>	Dec 2021
	· <u>Scaling Machine Learning for Remote Sensing using Cloud computing</u>	June 2021
	· <u>Global Flood Detection Challenge NASA-IMPACT</u> (137 participants)	April 2021 - June 2021
	· <u>Webinar</u> on building Deep Learning libraries (TensorFlow and PyTorch) from source on High Performance Computing (HPC) machines	Mar 2019
AWARDS & HONORS	· Appreciation for contribution to the GRSS Summer school on High-performance and Disruptive Computing in Remote Sensing	2021
	· Sharing is Caring Newcomer Award NASA IMPACT	2020
	· Smoky Mountains Computational Sciences & Engineering Conference (SMC) Data Challenge runner up	2018 and 2019
MENTORSHIP	Ilboudo Dieudonne (Burkina Faso), NeurIPS mentee	Aug 2024
	Ekansh Chauhan, IIIT Hyderabad MS student	Mar 2023 - May 2023
	Shuto Araki, DePauw University BS student	Jun 2019 - Aug 2019
	Emily Costa, FIU BS student	Jun 2019 - Aug 2019
SKILLS	Machine Learning: 5+ years in Generative AI, Classification, Regression, Segmentation, XGBoost, SVM, Decision Trees, Bayesian Inference, PCA, SVD, KDE, A/B testing, and geospatial data science. Programming: Proficient in Python (NumPy, Pandas, PyTorch, TensorFlow, etc.), SQL, JavaScript, Docker, NVIDIA Rapids, and AWS. Data Engineering: 4+ years in distributed ML and data engineering with tools like Ansible, Airflow, Dask, Ray, Kubernetes, and OpenMPI. Management: 5+ years leading teams and projects, including Rutogo, acquired by ixigo.com. (article)	